The Shape of the Tax

Round Prices and Bunching at New York City's 2019 Transfer-Tax Notches

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Abstract

New York's 2019 reform created buyer-paid transfer-tax notches at $2 million and above. Using 442,497 New York City sales, 2016–2025, I compare the same prices before and after the reform, against never-taxed round prices and the unchanged $1 million notch. About 60 percent of sales that had closed at exactly the round-number threshold close a few thousand dollars below it, whatever the size of the jump. Bunching estimators that assume a smooth price distribution read this as a large price response. Beyond the round price, responses are modest. Mortgage amounts bunch the same way below a $500,000 recording-tax notch.

JEL: H23, H71, K34, R31, R38. *Keywords:* transfer taxes; notches; bunching; round numbers; housing transactions; mortgage recording tax.

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1 Introduction

Housing transaction taxes are often notches: the rate applies to the whole price once a threshold is crossed, so one more dollar of price can carry thousands of dollars of tax. New York's 2019 budget created seven such notches at once. A buyer-paid supplemental tax took effect on 1 July 2019 at prices from \$2,000,000 to \$25,000,000, while the \$1,000,000 mansion tax, in place since 1989, did not change. The same thresholds can therefore be observed before and after they were taxed, next to an always-taxed control and never-taxed round prices. This paper uses that reform to ask how recorded prices respond to a notch, and what the answer implies for measuring such responses and for the shape of transaction taxes.

The answer is that the notches moved recorded prices, mostly by a little (Figure 1). Before the reform, \$2,000,000, \$3,000,000 and \$5,000,000 were ordinary round prices, each with a spike of sales at exactly that amount. After it, about 60% of those sales closed below the threshold instead, on average between \$2,000 and \$3,000 below it, less than the buyer's tax saving. The share is the same, within sampling error, for jumps of \$5,000, \$15,000 and \$37,500. Beyond the round price the response is modest: net of the unchanged \$1,000,000 notch and never-taxed round prices, sales in the 5% of prices above the threshold fall by about a fifth at \$3,000,000 and \$5,000,000 and by an amount indistinguishable from zero at \$2,000,000. Loan amounts show the same round-number pattern below the mortgage recording tax's \$500,000 loan notch.

The evidence comes from the 442,497 recorded residential sales above \$300,000 in New York City from 2016 to 2025. Three measures carry it. Counts of sales at exactly each threshold and just below it need no estimator (Tables 1 and 2). A difference-in-differences in counts by price zone nets out the always-taxed threshold and five never-taxed round prices (Table 3). And the bunching estimator of Kleven and Waseem (2013), applied before and after the reform at every threshold, places the response against placebo noise, across the whole schedule and year by year (Tables 4–6).

Two lessons reach beyond New York. First, at a round-number notch most of the measured bunching can be sales that would have closed at the round price closing a little below it. A bunching width computed against a smooth counterfactual reads them as a large price response: here a pile equivalent to \$16,500 of price at \$2,000,000, against a moved distance of about \$2,400. Notches in transaction taxes are almost always set at round prices, so their bunching needs a before period or an explicit treatment of the round price before it is read as a price response; the caution applies to old notches too, since three-quarters of the pile width at New York's \$1,000,000 notch lies within \$5,000 of the threshold. Second, that margin does not scale with the jump. With all sales counted, the response at

$5,000,000 matches the one at $2,000,000 although the jump is 7.5 times larger, and the shortfall from a response proportional to the rate disappears once the round-price sales are set aside. Whole-price notches are common: New Jersey's mansion tax, Washington, D.C.'s transfer taxes and the tax Los Angeles introduced in 2023 charge a higher rate on the whole price above a threshold (Kopczuk and Munroe 2015; Slemrod et al. 2017; Green et al. 2026), and the United Kingdom moved its stamp duty from whole-price to slice rates in December 2014 (Best and Kleven 2018; Kleven 2016). For New York, slice schedules that raise the same revenue as the 2019 notches without jumps are feasible but need higher marginal rates (Table 10).

The fiscal cost of the response is small. If the sales added below each threshold crossed it, as the price-zone counts indicate, they avoid between $1.3 and $2.7 million a year depending on how they are counted, under 1% of the $291 million a year that the reform's schedule implies on residential sales.

1.1 Relation to the literature

The notch evidence sits with Kopczuk and Munroe (2015), who study the New York and New Jersey transfer-tax notch and find both price bunching below the threshold and a shortfall of transactions above it that exceeds the bunching, so part of the market unravels instead of simply repricing. Best and Kleven (2018) document bunching at UK stamp-duty thresholds, Kleven and Waseem (2013) supply the notch estimator, Saez (2010) the kink design it extends, and Kleven (2016) a review of both. Best et al. (2020) use notches in UK mortgage rates at loan-to-value limits, and DeFusco and Paciorek (2017) estimate bunching of loan amounts at the conforming loan limit. Pope et al. (2015) show that negotiated house prices cluster at round numbers, which is the margin this paper finds the notches act on. Other natural experiments in transfer taxes point the same way. Slemrod et al. (2017) find sale prices moved to the lower-rate side of a notched change in Washington, D.C.'s transfer taxes, with many sales left in the dominated range. Dachis et al. (2012) find that Toronto's 1.1% land transfer tax reduced sales of single-family homes by 15% and lowered prices by about the amount of the tax, and Besley et al. (2014) that about sixty percent of the UK's 2008–09 stamp duty holiday went to buyers. Green et al. (2026) show that the transfer tax Los Angeles imposed in 2023 on sales above $5,000,000 sharply reduced transactions and eroded property-tax revenue through deferred reassessment; that channel is weak in New York City, where a sale does not reset a home's capped assessment (New York City Department of Finance 2018; Horton et al. 2024). Relative to these studies, the 2019 reform turns a fitted-counterfactual design into a before-and-after comparison at the same round

prices, and its seven thresholds on one date allow the response to be compared across jump sizes with the tax's age held fixed.

Section 2 sets out the tax schedules and Section 3 the data. Section 4 presents the evidence on recorded prices. Section 5 turns to the loan notch of the mortgage recording tax. Section 6 states what the paper cannot settle, and Section 7 concludes. The Supplemental Appendix holds the fixed-window estimates, the split by payment method, counts over equal periods, and the recording tax's gross charge on observed loans.

2 The tax schedules

2.1 The transfer-tax thresholds

The mansion tax under Tax Law §1402-a imposes 1% on residential conveyances of $1,000,000 or more. It is a notch, not a bracket: the tax applies to the whole consideration, so a sale at $1,000,000 owes $10,000 that a sale at $999,999 does not. The threshold has been in place, unchanged, since 1989.

The supplemental tax under §1402-b, on New York City residential conveyances, has seven brackets: 0.25% for consideration of $2,000,000 or more, 0.50% from $3,000,000, 1.25% from $5,000,000, 2.25% from $10,000,000, 2.50% from $15,000,000, 2.75% from $20,000,000 and 2.90% from $25,000,000. Each rate applies to the whole consideration, so every step is a notch, and the grantee pays it. It took effect for conveyances on or after 1 July 2019, with conveyances under binding written contracts entered into on or before 1 April 2019 excluded. The state base rate under §1402 rose in the same budget from 0.40% to 0.65% for residential conveyances of $3,000,000 or more, so the $3,000,000 threshold carries a half-point discontinuity: 0.25 points paid by the buyer and 0.25 points imposed on the seller. In sponsor (new-development) sales, contracts commonly shift the seller's transfer taxes to the buyer, so the statutory payer need not be the one who pays.

The city's own Real Property Transfer Tax steps from 1% to 1.425% on consideration above $500,000. Because a sale at exactly $500,000 already pays the lower rate, a price one dollar below avoids nothing there, and that threshold is not used in any inference (Supplemental Appendix Table A11).

The 2019 reform is the source of identification. It created seven discontinuities on one date, from $2,000,000 to $25,000,000, and left the $1,000,000 and $500,000 discontinuities untouched.

2.2 The mortgage recording tax

New York taxes the recording of a mortgage on real property (New York State 2026). For a one- to three-family home or a condominium unit in New York City the aggregate rate is 2.05% on mortgages below $500,000 and 2.175% at or above, on the whole loan. The special additional tax of 0.25% is payable by the lender for such properties under §253(1-a), so the borrower's rate is 1.80% below $500,000 and 1.925% at or above: a notch of $625 at the threshold. A purchase without a recorded mortgage pays none of it, and neither does a co-operative purchase, because a share loan is not a mortgage of real property. Supplemental Appendix Section I sets out the components and measures the gross charge on observed loans.

3 Data

Everything used here is public and free.

3.1 Transactions

The New York City Department of Finance publishes annualized calendar sales through NYC Open Data (resource `w2pb-icbu`), covering 1 January 2016 to 31 December 2025. The residential building classes are one-, two- and three-family dwellings, condominiums and co-operatives. With recorded consideration above $300,000 they give 442,497 transactions, of which 431,471 lie between $300,000 and $6,000,000. One formatting detail matters: calendar year 2016 writes the building-class category with a double space after the two-digit code, so a filter on the full literal strings silently drops the whole of 2016, the earliest pre-reform year. The filter used here matches on the two-digit code.

A deed that conveys several lots, such as a building sold whole or a unit with its storage lot, appears in the sales file once per lot, each row carrying the deed's whole consideration. Of the 442,497 rows, 32,894 share a sale date, borough and price with an earlier row; above $10,000,000 most rows belong to such groups. The main estimates use every row, as published. The difference-in-differences of Table 3 and the robustness checks of Section 4.5 also collapse such rows to one per deed, and the revenue calculations use collapsed rows throughout, because a multi-lot deed pays the transfer tax once.

3.2 Loans

New York City's Automated City Register Information System (ACRIS) records deeds and mortgages in Manhattan, the Bronx, Brooklyn and Queens; Staten Island instruments are recorded by the Richmond County Clerk. A purchase is classified as financed when a mortgage on the same parcel is dated within −15 to +90 days of the sale. Condominium units are matched to their unit lot through each sale's own deed, because the Department of Finance and ACRIS index condominiums under different lot numbers (Supplemental Appendix Section I). The result is a loan amount for 172,725 financed house and condominium purchases in the four boroughs: 125,023 houses and 47,702 condominiums.

4 Recorded prices at the 2019 notches

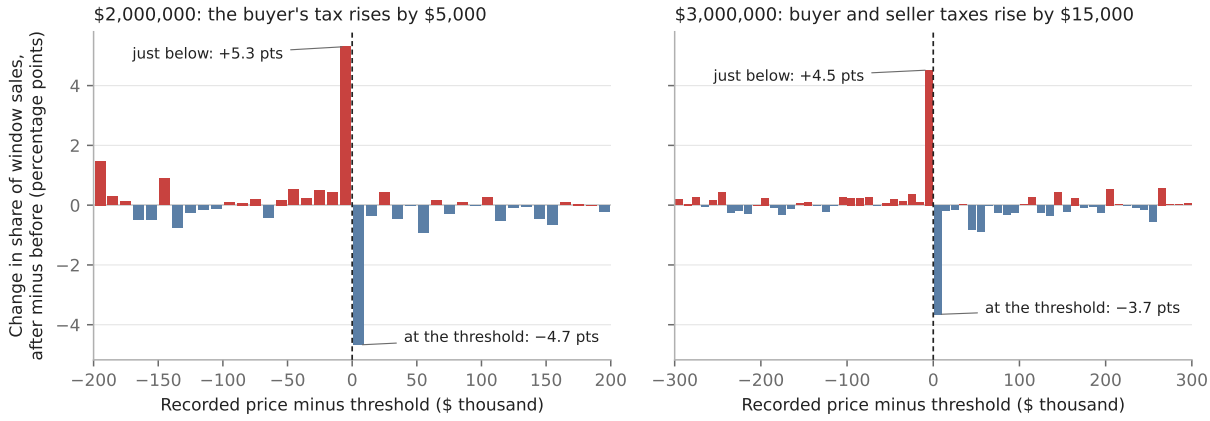


Figure 1: Recorded prices moved to just below the new thresholds. Each bar is the change in a $10,000 price bin's share of sales within 10% of the threshold, from January 2016–March 2019 to 2020–2025. The bin just below the threshold gains 5.3 percentage points at $2,000,000 and 4.5 at $3,000,000; the bin starting at the threshold, which includes sales at exactly the round price, loses 4.7 and 3.7. Prices are whole dollars, so the bin below the threshold includes sales at $X - 1$. Changes elsewhere, including at other round prices, are small. Replication package, `make_headline_figure.py`.

4.1 Design and measures

The 2019 budget created the notches effective 1 July 2019 and left the $1,000,000 mansion tax, in place since 1989, untouched. Conveyances under contracts signed on or before 1 April 2019 were grandfathered, so April to December 2019 is dropped as a transition window. The pre-period runs January 2016 to March 2019 (140,538 sales) and the post-period January

2020 to December 2025 (257,855 sales). Each threshold X is compared with itself before the reform, with the always-taxed \$1,000,000 threshold, and with never-taxed round prices. A price point whose estimation window contains no statutory threshold serves as a placebo. Four measures are used, from the most direct to the most model-based.

Counts by price zone. Let $s_{z,t}(X)$ be the share of sales in $[0.9X, 1.1X)$ that fall in price zone z in period t. The zones are $[0.95X, X - 5{,}000)$, $[X - 5{,}000, X)$, exactly X, and $(X, 1.05X)$, with the rest of the window as reference. The reform's change in zone z, in sales a year, is

$$\Delta_z(X) = \left[s_{z,\text{post}}(X) - s_{z,\text{pre}}(X)\right] \frac{N_{\text{post}}(X)}{6}, \tag{1}$$

where $N_{\text{post}}(X)$ counts post-reform sales in the window over six years, and the share of round-price sales that moved is $m(X) = 1 - s_{X,\text{post}}(X)/s_{X,\text{pre}}(X)$, with $z = X$ the zone of sales at exactly X.

A difference-in-differences in counts. Let n_{xtz} count sales in zone z around price point x in period t, for the three new thresholds, the \$1,000,000 control and five never-taxed \$100,000 multiples whose $\pm 10\%$ windows hold no threshold. The model is

$$\log \mathrm{E}[n_{xtz}] = \alpha_{xt} + \gamma_{xz} + \delta_{zt} + \beta_z \, \mathbf{1}[x \text{ new}] \, \mathbf{1}[t = \text{post}], \tag{2}$$

estimated by Poisson regression, with the two outer zones $[0.9X, 0.95X)$ and $[1.05X, 1.1X)$ as reference. Here α_{xt} absorbs each price point's volume in each period, γ_{xz} its fixed shape, including its own spike at the round price, and δ_{zt} any change in shape common to all nine price points. Then $\exp(\beta_z) - 1$ is the change in zone z at the new thresholds net of the same change at the control and the placebos. Intervals come from a block bootstrap over calendar months.

Bunching. The estimator follows Kleven and Waseem (2013) and Kopczuk and Munroe (2015). In bins j of width $0.005X$ (rounded down to \$10,000, \$25,000, \$50,000 or \$100,000) within $\pm 0.2X$, counts outside an excluded range $(X - 0.025X, X + 0.05X)$ are fitted by

$$c_j = \sum_{k=0}^{5} \theta_k \left(\frac{p_j - X}{0.2X}\right)^k + \rho' r_j + u_j, \tag{3}$$

where p_j is the bin's midpoint and r_j marks round-number bins. The statistic is the share of counterfactual mass missing above the threshold, $b(X) = \sum_{j \in A}(\hat{c}_j - c_j) / \sum_{j \in A} \hat{c}_j$ over the bins A in $(X, 1.05X)$, and the reported change is $b_{\text{post}} - b_{\text{pre}}$. As implemented, r_j marks

the two $10,000 bins on either side of each multiple of $100,000; marking the bins that contain multiples of $100,000, $50,000 and $25,000 instead changes the estimates by at most 0.017 (Section 4.5). Bootstrap intervals resample sales within the window in each period. A threshold with fewer than 1,000 pre-reform sales in its window is called thin.

A ratio with no counterfactual. $R(X) = N[X, 1.05X)/N[0.925X, 0.975X)$ compares sales just above the threshold with sales just below it, outside the range where bunching sits; a notch predicts a fall in $\log R$.

4.2 The round price

Round numbers attract transactions (Pope et al. 2015). A notch at a round number makes it the first taxed price, so a sale that would have closed there can avoid the jump by recording a slightly lower price. Table 1 counts this directly. Before the reform, $2,000,000 and $3,000,000 carried 7.8 and 7.0 times the surrounding density of sales at exactly that price, as high as the never-taxed $2,500,000. After it the spikes fell to 1.6 and 1.5, while those at $1,500,000 and $2,500,000 rose. Sales one dollar below went from 4 to 131 and from 2 to 68; at the never-taxed prices they stayed at four or fewer. At $1,000,000, a notch since 1989, there is no spike at exactly the threshold in either period, and sales one dollar below are already common.

Table 1: Focal prices, before and after the 2019 reform

	Spike at exactly X, relative to local density		Sales at $X - 1$	
Price point	Before Apr 2019	From Jan 2020	Before	From Jan 2020
$1,000,000 — taxed throughout	0.1	0.1	156	543
$2,000,000 — treated	**7.8**	**1.6**	**4**	**131**
$3,000,000 — treated	**7.0**	**1.5**	**2**	**68**
$1,500,000 — never taxed	5.2	6.2	3	4
$2,500,000 — never taxed	7.8	10.6	1	0

Notes. The two periods are **before 1 April 2019** and **from 1 January 2020**, leaving out the nine months in between: conveyances under binding contracts signed on or before 1 April 2019 were grandfathered, so the intervening sales are a mixture of treated and untreated and belong to neither period. This requires no estimator, no counterfactual and no distributional assumption. It is a count of transactions at five prices.

Table 2 follows the moved sales by price zone, using equation (1). At $2,000,000 the reform added about 90 sales a year in the $5,000 below the threshold and 36 between

$1,900,000 and $1,995,000. It removed 77 a year at exactly $2,000,000 and 31 from the rest of the 5% above. Of the sales missing above the threshold, 71% were at the round price itself; at $3,000,000 and $5,000,000 the share is 57% and 49%. Of the sales that used to close at exactly the threshold, 58% moved at $2,000,000, 60% at $3,000,000 and 62% at $5,000,000, with 95% intervals of 51–64%, 51–69% and 47–74%; equality of the three is not rejected ($p = 0.88$), although the total jumps are $5,000, $15,000 and $37,500. The never-taxed round prices' own spikes grew by 13% on average, so against them the moved shares are slightly larger, 63% to 66%. The sales added just below sit on average $2,400 below $2,000,000, $2,200 below $3,000,000 and $2,900 below $5,000,000, less than the buyer's saving at each threshold. The never-taxed round prices show none of this. The always-taxed $1,000,000 threshold also gained sales just below it, 140 a year, partly (50 a year) from the 5% above; its tax did not change, and the next subsection nets that drift out.

The remaining 40% of round-price sales still close at exactly the threshold and pay the higher tax on the whole price for one extra dollar, whatever the size of the jump. A fixed cost of changing a price agreed at a round number, or inattention to the threshold, would produce that pattern; these data cannot tell the two apart. Slemrod et al. (2017) read sales left in the dominated range of Washington, D.C.'s notches the same way.

4.3 Netting out the control and never-taxed round prices

Table 3 estimates equation (2). Pooled over the three new thresholds, sales at exactly the threshold fall by 61% relative to the outer zones and net of the control and the placebos, and sales in the $5,000 below it more than triple. Sales between $0.95X$ and $X - 5{,}000$ rise by about a tenth, and sales in the 5% above the threshold, excluding the round price, fall by 10% with all rows (interval −20% to 0%) and by 14% with repeated deed rows collapsed (−20% to −8%). The last line is the response beyond the round price. By threshold, on collapsed rows, it is −6% at $2,000,000, with an interval from −17% to +4%, and −20% and −24% at $3,000,000 and $5,000,000, each interval excluding zero. Using the control alone or the placebos alone as the comparison gives similar changes at the round price (−52% and −62%) and above it (−9% and −16%).

4.4 Bunching across the schedule

The bunching estimator puts the same response on a common scale at every threshold. Table 4 reports it for the seven thresholds created in 2019 and for never-taxed round prices whose windows hold no threshold. At $2,000,000, $3,000,000 and $5,000,000 the share of counterfactual mass missing above the threshold rises by 0.220, 0.321 and 0.219, and $\log R$

Table 2: Where the moved sales came from

Threshold	Change in sales a year, by recorded price: $[0.95X, X-5{,}000)$	$[X-5{,}000, X)$	Exactly X	$(X, 1.05X)$	Mean distance below X of added sales	Sales at X that moved
\$2,000,000	+36	+90	−77	−31	\$2,396	58%
	[+9, +63]	[+79, +102]	[−92, −62]	[−55, −9]	[\$1,933, \$2,859]	[51, 64]
\$3,000,000	+21	+42	−37	−28	\$2,196	60%
	[+1, +42]	[+34, +50]	[−47, −28]	[−45, −11]	[\$1,525, \$2,724]	[51, 69]
\$5,000,000	+14	+23	−16	−16	\$2,906	62%
	[−2, +29]	[+19, +28]	[−23, −9]	[−28, −3]	[\$2,246, \$3,529]	[47, 74]
\$1,000,000 (control)	−31	+140	+4	−50	\$2,611	—
	[−80, +26]	[+108, +179]	[−2, +10]	[−75, −25]	[\$1,370, \$3,609]	
\$1,500,000 (never taxed)	+13	−8	+8	−4	—	—
	[−17, +48]	[−19, +3]	[−13, +28]	[−35, +27]		
\$2,500,000 (never taxed)	−22	−3	−1	+1	—	—
	[−45, +1]	[−9, +4]	[−14, +12]	[−19, +24]		

Notes. Each cell is the change in a price zone's share of sales between $0.9X$ and $1.1X$, from January 2016–March 2019 to 2020–2025, multiplied by post-reform sales a year in that range: the number of sales a year the zone gained or lost. The mean distance is the mean of $X - P$ over the sales added to $[X - 10{,}000, X)$: post-reform sales there, less pre-reform sales rescaled to the post-reform window size. The share that moved is one minus the ratio of the post-reform to the pre-reform share of window sales recorded at exactly X; equality across the three thresholds is not rejected ($p = 0.88$). Prices are whole dollars, so the zone just below X includes $X - 1$. 95% bootstrap intervals (500 draws; 1,000 for the share that moved), resampling sales within the window in each period, in brackets. Replication package, `focal_robustness.py`, `focal_inference.py`.

falls by 0.328, 0.348 and 0.466; sales at \$4,999,999 go from none to 26. Every placebo has an interval that includes zero on the bunching measure. Above \$5,000,000 the thresholds are thin and the estimator returns large values of both signs. The simplest count, sales in the 1% of price below each threshold against the 1% above, still reads: on rows collapsed to one per deed, it shifts by +1.93 log units at \$10,000,000 and +1.32 at \$15,000,000, against +1.66 at \$2,000,000, but on fewer than 30 pre-reform sales in each band (Supplemental Appendix Table A3).

With all sales, the response is not proportional to the rate (Figure 2). The point estimate at \$5,000,000, where the buyer's rate rises by 0.75 points and the jump is \$37,500, matches the one at \$2,000,000, where it rises by 0.25 points and the jump is \$5,000: bootstrapping the two jointly, the difference is −0.001 [−0.190, +0.219]. A response proportional to the buyer's rate would make the \$5,000,000 change three times the \$2,000,000 one; the shortfall is −0.441 [−0.769, −0.118] ($p = 0.008$). Adding the seller's 0.25 points at \$3,000,000 raises the response by +0.101 [−0.064, +0.278], and equal responses at all three thresholds are not rejected ($p = 0.53$). Section 4.5 shows that the shortfall from proportionality is the

Table 3: Bin-count difference-in-differences: the new thresholds against the control and never-taxed round prices

	Pooled, $2M, $3M, $5M		By threshold, rows collapsed		
Price zone	All rows	Rows collapsed	$2,000,000	$3,000,000	$5,000,000
$[0.95X, X - 5{,}000)$	+11	+10	+12	+8	+10
	[+3, +20]	[+2, +18]	[+1, +23]	[−3, +20]	[−5, +29]
$[X - 5{,}000, X)$	+229	+239	+229	+188	+584
	[+176, +289]	[+182, +312]	[+169, +306]	[+124, +293]	[+307, +1568]
Exactly X	−61	−62	−60	−66	−63
	[−68, −53]	[−67, −56]	[−66, −52]	[−73, −56]	[−76, −45]
$(X, 1.05X)$	−10	−14	−6	−20	−24
	[−20, +0]	[−20, −8]	[−17, +4]	[−30, −9]	[−36, −12]

Notes. Percentage change in the number of sales in each zone after the reform (2020–2025 against January 2016–March 2019), relative to the two outer zones $[0.9X, 0.95X)$ and $[1.05X, 1.1X)$ and net of the same change at the always-taxed $1,000,000 threshold and at five never-taxed round prices ($1.5, 2.5, 3.5, 4 and 4.5 million, whose ±10% windows hold no threshold). Poisson regression on zone counts with price point × period, price point × zone and zone × period effects; reported as $\exp(\beta) - 1$. 95% intervals from 500 block-bootstrap draws that resample calendar months within each period. Collapsed rows keep one row per sale date, borough and price (Section 3). With the control alone as comparison, the pooled changes at exactly X and in $(X, 1.05X)$ are −52% and −9%; with the placebos alone, −62% and −16%. Replication package, `focal_inference.py`.

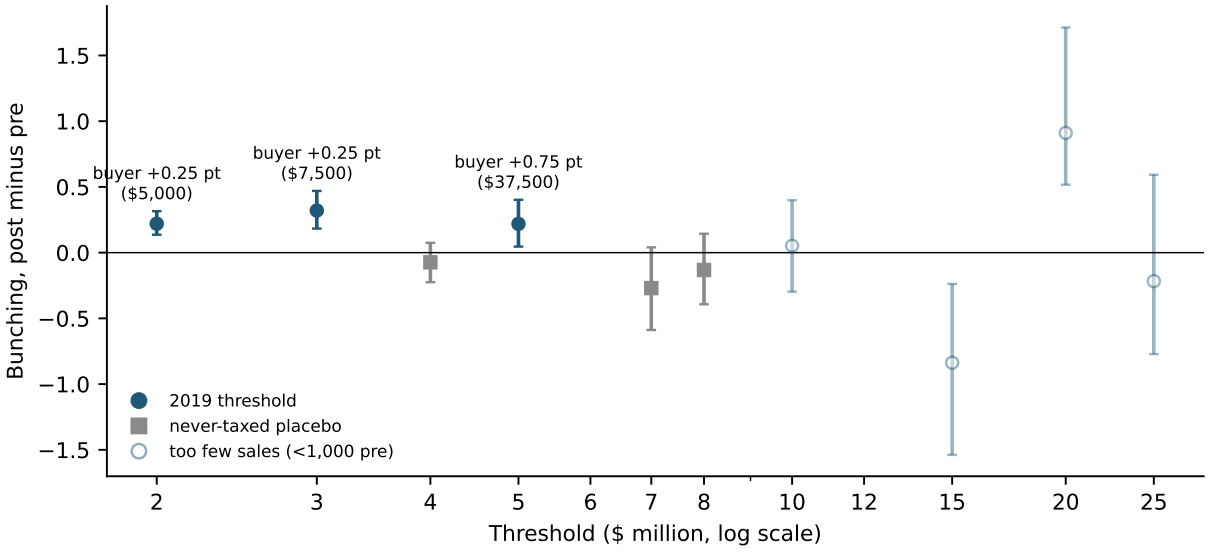


Figure 2: With all sales, the response to a buyer-paid notch does not rise in proportion to the buyer's jump. Change in bunching after the 2019 reform at each threshold of the schedule, with 95% bootstrap intervals (Table 4). Without the sales at the threshold and one dollar below it, proportionality is not rejected (Table 6). Filled circles are reformed thresholds with enough sales to estimate; squares are never-taxed placebos whose windows hold no threshold; hollow markers have fewer than 1,000 pre-reform sales in the window.

Table 4: The full 2019 schedule: every threshold, before and after the reform

Threshold	Buyer's jump (rate / dollars)	Sales in window pre/post	Bunching, post − pre [95% CI]	$\Delta \log R$, no counterfactual [95% CI]	Sales at $X-1$ pre → post
Panel A. Thresholds created by the 2019 reform					
\$2,000,000	0.25 pt / \$5,000	9,299/23,022	+0.220 [+0.136, +0.316]	−0.328 [−0.434, −0.223]	4 → 131
\$3,000,000	0.25 pt / \$7,500[a]	5,268/12,374	+0.321 [+0.183, +0.469]	−0.348 [−0.483, −0.214]	2 → 68
\$5,000,000	0.75 pt / \$37,500	2,392/5,791	+0.219 [+0.046, +0.402]	−0.466 [−0.663, −0.270]	0 → 26
\$10,000,000[b]	1.00 pt / \$100,000	817/1,892	+0.052 [−0.297, +0.399]	−0.258 [−0.622, +0.106]	3 → 7
\$15,000,000[b]	0.25 pt / \$37,500	415/1,207	−0.838 [−1.539, −0.238]	+0.198 [−0.235, +0.631]	0 → 2
\$20,000,000[b]	0.25 pt / \$50,000	221/1,186	+0.911 [+0.517, +1.713]	−2.198 [−2.812, −1.584]	0 → 0
\$25,000,000[b]	0.15 pt / \$37,500	172/647	−0.217 [−0.772, +0.592]	−1.124 [−1.895, −0.353]	0 → 0
Panel B. Round-number placebos, never taxed					
\$4,000,000	none	3,348/7,753	−0.074 [−0.225, +0.075]	+0.036 [−0.126, +0.198]	1 → 5
\$7,000,000	none	1,509/3,550	−0.270 [−0.588, +0.040]	+0.010 [−0.224, +0.245]	0 → 0
\$8,000,000	none	1,234/2,849	−0.132 [−0.393, +0.144]	+0.275 [+0.007, +0.544]	0 → 0

Notes. Pre is January 2016 to March 2019; post is January 2020 to December 2025. Windows scale with the threshold so one design applies across the schedule. Placebos are never-taxed round prices whose window contains no threshold. Bunching is the Kleven–Waseem share of counterfactual mass missing above X (window $\pm 0.2X$, bins of $0.005X$ rounded down to \$10,000, \$25,000, \$50,000 or \$100,000, exclusion band $(-0.025X, +0.05X)$, degree-five polynomial with round-number indicators), post minus pre, with bootstrap percentile intervals. R is sales in $[X, 1.05X)$ over sales in $[0.925X, 0.975X)$; its log change has a Poisson interval, so intervals differ slightly from the bootstrap intervals of the fixed-window design (Supplemental Appendix Table A10). [a]At \$3,000,000 the seller's base rate also rises 0.25 points; the fixed ±\$400,000 window gives +0.461 (Supplemental Appendix Table A10). [b]Fewer than 1,000 pre-reform sales in the estimation window. On samples this small the estimator returns large, individually significant values of both signs at treated thresholds (\$15,000,000 and \$20,000,000), so no conclusion is drawn from these rows.

round-price margin.

Inference against never-taxed prices. Table 5 judges each estimate against the same statistic at every \$50,000 multiple between \$1,200,000 and \$6,000,000 whose window contains no threshold: sixteen prices, from \$1,300,000 to \$1,650,000 and from \$3,800,000 to \$4,150,000. Their bunching changes average −0.009, with a standard deviation of 0.047 and a maximum of +0.080. The changes at the three new thresholds are each larger than all sixteen, and each log change in R is lower than all sixteen. Neighboring windows share sales, so the rank $(1 + k)/17$, where k counts placebos at least as extreme, is a reference comparison rather than a calibrated p-value. Resampling each larger placebo window to a treated threshold's own sample size, the \$2,000,000 and \$3,000,000 changes lie 3.7 and 4.3 standard deviations from the draws' mean and exceed every draw; the \$5,000,000 change, on a quarter of the sample, lies 1.9 standard deviations out, with 4% of draws as large. Against these draws the shortfall from proportionality is less unusual: 4.8% of placebo contrasts are at least as

extreme (5.3% when the two placebo windows must not overlap), and 33% against a wider set of 41 thinner placebos up to $8,300,000. The control does not stand out on bunching, but its log change in R, -0.095, is lower than every placebo; the event study locates that change.

Table 5: Placebo-calibrated inference for the 2019 notches

	Change,	Placebos at	Rank	Size-matched placebos		
Threshold	post − pre	least as extreme	$(1+k)/17$	Resampling SD	z	As large
Panel A. Bunching (share of counterfactual mass missing above X)						
$1,000,000 (control)	+0.055	2 of 16	0.18	—	—	—
$2,000,000	+0.220	0 of 16	0.06	0.062	3.7	0 of 400
$3,000,000	+0.321	0 of 16	0.06	0.078	4.3	0 of 400
$5,000,000	+0.219	0 of 16	0.06	0.121	1.9	31 of 800
Panel B. Log change in R, no counterfactual fitted						
$1,000,000 (control)	−0.095	0 of 16	0.06	—	—	—
$2,000,000	−0.328	0 of 16	0.06	—	—	—
$3,000,000	−0.348	0 of 16	0.06	—	—	—
$5,000,000	−0.466	0 of 16	0.06	—	—	—
Panel C. Tests across thresholds (bunching)						
	Estimate	Bootstrap p		Size-matched placebo share		
$5M − $2M	−0.001	0.99		0.99		
$5M − 3 × $2M (proportional)	−0.441	0.008		0.048		
$3M − $2M	+0.101	0.28		0.31		

Notes. Estimator and windows as in Table 4. Placebos are the sixteen $50,000 multiples between $1,200,000 and $6,000,000 whose window $[0.8X, 1.2X)$ contains none of the thresholds at $500,000, $1,000,000, $2,000,000, $3,000,000, $5,000,000 and above. The rank column is one-sided in the direction a notch predicts (bunching up, R down): $(1 + k)/17$, where k is the number of placebos at least as extreme. It is a reference ranking rather than an exact p-value: a calibrated rank test would need an exchangeability argument the design does not supply, and neighboring placebo windows overlap. Size-matched placebos resample each placebo window with more sales than the treated window, with replacement, to the treated window's pre- and post-reform counts, fifty times per placebo (eight placebos for $2,000,000 and $3,000,000, sixteen for $5,000,000); z is the estimate's distance from the draws' mean in their standard deviations. No placebo has as many sales as the $1,000,000 window, so the control has no size-matched row. Panel C reports the two-sided share of differences between draws at two distinct placebos that are at least as large (for the proportional contrast, 0.053 when the two placebo windows must not overlap); these shares are reference comparisons, not calibrated p-values. Bootstrap p-values are from the joint bootstrap of Section 4.4. Supplemental Appendix Table A6 lists every placebo. Replication package, `placebo_inference.py`.

Timing. Figure 3 computes both statistics year by year, each normalized to its own 2016–2018 mean, against the 90% range of the same series at size-matched placebos. Before

the reform the treated series stay inside the band in every year. From 2020 the change at \$2,000,000 is immediate and does not fade: log R lies below the band in all six post-reform years, between -0.27 and -0.35, and bunching lies above it in five. At \$3,000,000 log R is outside the band in five of six years and bunching in four; at \$5,000,000, the smallest sample, in four and one. Moving the reform back to January 2018 gives bunching changes of -0.112, -0.045 and -0.022, all inside their placebo distributions (shares of 0.53 or more). The control is flat on bunching, apart from two excursions of less than 0.03 beyond the band, but its log R falls in 2020–2022, by 0.33 in 2020, and is back near its baseline from 2023 (just above the band that year, inside it in 2024 and 2025). The dip coincides with the pandemic. From 2023 to 2025, with the control at its baseline, log R stays between -0.27 and -0.35 at \$2,000,000, between -0.29 and -0.55 at \$3,000,000 and between -0.31 and -0.97 at \$5,000,000. The later estimates at the two higher thresholds are larger than the early ones, which fits a response that grows as a notch becomes familiar, but the yearly samples are too small to establish a trend; at \$2,000,000 there is no such pattern.



Figure 3: The response starts with the reform. Bunching (top) and log R (bottom) by year, each normalized to its 2016–2018 mean, at the always-treated \$1,000,000 threshold and the three thresholds created in 2019. Grey bands are the 90% range of the same statistic at never-taxed placebos resampled with replacement to each threshold's yearly sales (for the control, the seven placebos with at least half its sales, without subsampling). Hollow points mark 2019, the transition year, which no test uses. Replication package, `event_study.py`.

4.5 Beyond the round price, and other robustness

Table 6 re-estimates the bunching statistics without the round-price sales, at the thresholds and at every placebo alike. Dropping only the sales at exactly X and $X-1$ cuts the change at \$2,000,000 from 0.220 to 0.044 [−0.027, +0.120], exceeded by two of the sixteen placebos and below the control's 0.058; dropping everything within \$1,000 or \$5,000 of the threshold gives the same answer. At \$2,000,000 the measured bunching is therefore almost entirely the round price. At \$3,000,000, 0.146 [+0.027, +0.262] remains, above every placebo, and about 0.17 when sales within \$1,000 or \$5,000 are dropped; at \$5,000,000 about 0.10 remains, exceeded by one placebo, with an interval that includes zero. Net of the control's own change neither is significant (+0.088 [−0.028, +0.202] and +0.043 [−0.093, +0.197]). Among houses, whose control is flat, the change at \$3,000,000 is +0.235, exceeded by one placebo, and with repeated deed rows collapsed the changes are +0.206 and +0.177, above every placebo. The response beyond the round price at the two higher thresholds is therefore present in the count-based design of Table 3 and in some but not all bunching specifications; at \$2,000,000 it is absent in all of them.

Without the round-price sales, the test of proportionality loses its force. The \$5,000,000 change is close to three times the \$2,000,000 one (−0.031, $p = 0.82$), but equality is not rejected either ($p = 0.46$), so the test has little power. What the full-sample rejection reflects is the round-price margin, whose size is set by how many deals sit at the round number and which, at about 60%, did not vary with the jump.

Other checks. Marking the round-number bins that contain multiples of \$100,000, \$50,000 and \$25,000, rather than the bins on either side of each multiple of \$100,000, changes the all-sales and round-price-excluded estimates by at most 0.017. A fixed ±\$400,000 window with \$10,000 bins gives changes of 0.220 and 0.461 at \$2,000,000 and \$3,000,000 and +0.005 at the control, stable across period definitions (Supplemental Appendix Tables A1 and A10); the control's change is larger in the scaled window used here, which is why the comparisons above net it out. All estimates use nominal consideration, to which the statute attaches; the share of sales above \$2,000,000 rose from 8.4% to 12.3% over the decade, but a 2017–18 against 2020–21 comparison, over which prices changed little, leaves the \$2,000,000 response at +0.202 in the fixed window. In the pre-period the estimator returns a missing share of 0.094 at the then-untaxed \$2,000,000, with an interval excluding zero: levels are biased, which is why only differences are used and every estimate is judged against placebos.

Table 6: The response with and without the round price

Panel A. Change in bunching, post − pre (placebos at least as large, of 16)				
	\$1,000,000 (control)	\$2,000,000	\$3,000,000	\$5,000,000
All sales (Table 4)	+0.055 (2)	+0.220 (0)	+0.321 (0)	+0.219 (0)
Without sales at X and $X-1$	+0.058 (2)	+0.044 (2)	+0.146 (0)	+0.101 (1)
	[+0.027, +0.084]	[−0.027, +0.120]	[+0.027, +0.262]	[−0.034, +0.263]
Without sales within \$1,000	+0.060 (2)	+0.037 (3)	+0.169 (0)	+0.101 (1)
Without sales within \$5,000	+0.070 (1)	+0.039 (3)	+0.172 (0)	+0.107 (1)
Without sales at X and $X-1$:				
Net of the control	—	−0.014	+0.088	+0.043
		[−0.091, +0.069]	[−0.028, +0.202]	[−0.093, +0.197]
Houses only	+0.005 (6)	+0.039 (4)	+0.235 (1)	+0.120 (3)
Repeated rows collapsed	+0.055 (5)	+0.039 (5)	+0.206 (0)	+0.177 (0)
Indicators on containing bin	+0.064	+0.042	+0.138	+0.091
Panel B. Log change in R (placebos at least as low, of 16)				
All sales	−0.095 (0)	−0.328 (0)	−0.348 (0)	−0.466 (0)
Without sales at X and $X-1$	−0.114 (0)	−0.127 (0)	−0.199 (0)	−0.348 (0)

Panel C. Tests across thresholds (bunching)				
	All sales		Without X and $X-1$	
	Estimate	Bootstrap p	Estimate	Bootstrap p
\$5M − 3 × \$2M (proportional)	−0.441	0.008	−0.031	0.82
\$5M − \$2M	−0.001	0.990	+0.057	0.46
\$3M − \$2M	+0.101	0.280	+0.102	0.15

Notes. Estimator, windows, periods and placebos as in Tables 4 and 5. Each row drops the stated sales in both periods, at the thresholds and at every placebo alike; the dropped prices lie inside the range left out of the counterfactual fit, and where a bin loses part of its price range its counterfactual count is scaled by the share kept. 95% bootstrap intervals (500 draws) in brackets. Net of the control subtracts the \$1,000,000 change, with independent draws for the two windows, which do not overlap. Houses are building classes 01–03, with placebos recomputed among houses. Repeated deed rows: a deed that conveys several lots appears once per lot in the sales file at the whole consideration; collapsing rows that share a sale date, borough and price removes 32,894 of 442,497 rows. The published estimator marks, for \$10,000 bins, the two bins on either side of each multiple of \$100,000 (Section 4.5); the last row of Panel A marks instead the bin that contains each multiple of \$100,000, \$50,000 and \$25,000. R is sales in $[X, 1.05X)$ over sales in $[0.925X, 0.975X)$. Panel C: all-sales p-values from 1,000 joint draws (Section 4.4); without X and $X-1$, from 500. Replication package, `focal_robustness.py`.

4.6 Where the response comes from

The linked dataset (Loschi 2026a) attaches property type, borough, the seller on the deed and the deed's classification to the same sales. Table 7 splits the bunching change by each, against placebos recomputed within the same sample. Among houses the changes at the three new thresholds are +0.281, +0.487 and +0.502, and among condominiums +0.206, +0.339 and +0.183; in Manhattan +0.223, +0.324 and +0.165, and in the other four boroughs +0.206, +0.370 and +0.602. The house estimates at $3,000,000 and $5,000,000 rest on 1,012 and 303 pre-reform sales, and the last outer-borough estimate on 342, so they are imprecise. The control's movement in the pooled data comes from condominiums (+0.133) and Manhattan (+0.095); among houses it does not move (+0.008, with eight placebos as large). Among condominiums the new thresholds' changes exceed their own control's by only +0.07, +0.21 and +0.05, so the condominium evidence rests mainly on $3,000,000. Co-operatives' changes, +0.174, +0.161 and +0.086, are each matched by two placebos in a noisy sample.

Developers set condominium prices in advance, and some sponsor sales that closed after the reform were contracted before the April 2019 cut-off. The linked deed shows whether a company sold, a proxy that includes sponsor sales. In sales without a company seller the changes are +0.349, +0.384 and +0.511, each larger than every placebo in that sample, and the control does not move (+0.022). In company-seller sales the response is clear at $3,000,000 (+0.431) and within placebo noise at $2,000,000 (Supplemental Appendix Table A8). Referee's deeds and deeds in lieu of foreclosure are recorded as plain deeds and stay in the sales file, but they are rare near these prices (0.2–1.2% of window sales have a lender or referee party); dropping them leaves the changes at +0.225, +0.324 and +0.222.

4.7 How far prices moved, how the deals were struck, and what it cost

How far. Dividing the excess mass below a threshold and the missing mass above it by the counterfactual density at the threshold expresses each as a width of price (Table 8). With all sales, the pile below $2,000,000 grew by the equivalent of $16,500 of price, more than three times the $5,000 jump. Read literally, the moved sales cut their prices by far more than the tax they saved. Table 2 shows why that reading is wrong: the added sales sit on average $2,400 below the threshold, and most came from the round price itself. A width measures how much price range a pile would empty if its sales came from a smooth distribution; when they come from a mass point at the threshold, it overstates the price change. Without sales within $5,000 of each threshold, the pile at $2,000,000 shrinks to $5,100 and the hole above it to $800, and at $3,000,000 and $5,000,000 the holes shrink to $10,900 and $19,400, each

Table 7: Where the response comes from

Change in bunching, post − pre (placebos at least as large, of 16)					
	\$1,000,000 (control)	\$2,000,000	\$3,000,000	\$5,000,000	Placebo SD
All sales	+0.055 (2)	+0.220 (0)	+0.321 (0)	+0.219 (0)	0.047
Houses	+0.008 (8)	+0.281 (2)	+0.487 (0)	+0.502 (0)	0.152
Condominiums	+0.133 (1)	+0.206 (0)	+0.339 (0)	+0.183 (0)	0.083
Co-operatives	+0.027 (4)	+0.174 (2)	+0.161 (2)	+0.086 (2)	0.211
Manhattan	+0.095 (0)	+0.223 (0)	+0.324 (0)	+0.165 (0)	0.061
Other four boroughs	+0.033 (3)	+0.206 (0)	+0.370 (0)	+0.602 (0)	0.138
No company seller	+0.022 (6)	+0.349 (0)	+0.384 (0)	+0.511 (0)	0.067
Company seller (sponsor proxy)	+0.090 (6)	+0.067 (6)	+0.431 (0)	+0.080 (6)	0.181
Without legal-rule deeds	+0.050 (2)	+0.224 (0)	+0.325 (0)	+0.219 (0)	0.048
Without any lender or referee deed	+0.050 (2)	+0.225 (0)	+0.324 (0)	+0.222 (0)	0.048

Notes. Estimator, windows and periods as in Table 4. Property types follow the Department of Finance building-class category (houses 01–03; condominiums 04, 12, 13, 15; co-operatives 09, 10, 17) and come from the linked dataset (Loschi 2026a), whose sales are the same records as this paper's sample. Each row's placebos are the sixteen of Table 5, recomputed within that row's sample. The last two market rows split sales by whether the linked deed names a company as seller, a proxy that includes sponsor (developer) sales and other company sales; sales without a linked deed are in neither row. Deed screens use the linked dataset's deed classifier (Loschi 2026b); sales with no linked deed (co-operatives, Staten Island sales, and houses and condominiums whose deed was not matched) are kept. Supplemental Appendix Table A7 adds log changes in R and window sizes. Replication package, `market_splits.py` and `cost_and_sponsors.py`.

narrower than the dominated range, the band above the threshold in which a buyer who noticed the notch would never choose to transact. Many sales therefore stay where the notch leaves them strictly worse off, the signature of optimization frictions (Kleven and Waseem 2013). The long-standing \$1,000,000 notch looks the same: three-quarters of its post-reform pile width lies within \$5,000 of the threshold.

The usual parameters inherit the problem. The local incidence of Kopczuk and Munroe (2015), the pile width relative to the jump, is 3.30 at \$2,000,000 with all sales; for the sales that moved off the round price, the recorded price fell by about half the buyer's \$5,000 saving at \$2,000,000 and by less than a fifth of the jump at \$3,000,000 and \$5,000,000. Their unraveling statistic, the hole width less the pile width, is not informative here: its interval includes zero at the new thresholds, and at \$1,000,000 it is significantly negative with this paper's estimator (−\$4,658) and significantly positive with Kopczuk and Munroe's level-shift counterfactual (\$26,100). Read through the frictionless model of Kleven and Waseem (2013), with $1 + t$ in place of $1 - t$, the all-sales widths imply recorded-price elasticities of 0.005 to

0.016 at the new notches and 0.04 to 0.05 at $1,000,000; without the sales within $5,000 of each threshold, those at the new notches fall to 0.002 or less. Because the round-price margin dominates the widths and frictions dominate beyond it, these numbers summarize local responses in recorded prices rather than structural elasticities, and the gap between new and old notches cannot be read as an effect of a notch's age.

Table 8: How far sales moved: bunching widths with and without the round price

	Jump	Dominated	Pile width		Hole width		Elasticity,	
	T	range	All	Without	All	Without	all sales	
Threshold			($ thousand)				Pile	Hole
$2,000,000	5.0	4.9	16.5	5.1	10.7	0.8	0.016	0.008
$3,000,000	15.0	14.5	16.7	3.8	21.9	10.9	0.005	0.008
$5,000,000	37.5	35.9	32.5	−5.5	43.7	19.4	0.005	0.008
$1,000,000 (level)	10.0	—	30.0	7.7	25.3	21.4	0.053	0.039

Notes. Widths are the excess mass in $(0.975X, X)$ (pile) and the missing mass in $(X, 1.05X)$ (hole), each divided by the counterfactual density of sales per dollar at X; at the 2019 thresholds they are post-reform minus pre-reform, at $1,000,000 the post-reform level. "Without" drops sales within $5,000 of X in both periods, as in Table 6. T is the jump in all transfer taxes at X; the dominated range is the band above X in which a buyer who saw the notch could never prefer to transact, $T/(1+t)$ with t the total transfer-tax rate just above X. The elasticity is that of the recorded price with respect to one plus the transfer-tax rate, solved from the indifference condition of the marginal buncher (Kleven and Waseem 2013) with $1+t$ in place of $1-t$, all transfer taxes counted and no optimization frictions, from the all-sales widths; counting only buyer-paid taxes changes them by at most 0.005. Replication package, `focal_robustness.py`, `notch_elasticity.py`.

How the deals were struck. A sale recorded at $1,997,600 may have been negotiated at that price, or agreed at $2,000,000 with value moved off the deed into personal property, credits or closing adjustments. The records cannot observe side payments, but loan amounts give one check. If deals were agreed at the round price and trimmed afterwards, loans sized on that price would often be round fractions of it. Among 353 financed post-reform purchases recorded in the $10,000 below $2,000,000, 10% have a loan equal to 50%, 55%, ..., or 95% of $2,000,000, the same as before the reform (8% of 37), while 43% have a loan equal to one of those fractions of the recorded price instead. At exactly $2,000,000 the two coincide, and 71% of loans are such fractions. At $3,000,000 the shares are 7% and 43% of 156. Most moved sales were therefore priced below the threshold by the time the loan was sized. This rules out a late trim but not consideration moved off the deed during negotiation; Montalvo et al. (2020) propose comparing declared prices with appraisals, which are not in the public record.

What the notches cost. A sale that moves below a notch avoids at least the jump in tax. With the published estimator, about 130 extra sales a year sit just below $2,000,000, 54 below $3,000,000 and 20 below $5,000,000 (Table 9), each pile larger as a share of its window than at any placebo. If they crossed the threshold, as Table 2 indicates, they avoid $2.2 million a year at the jump alone (95% interval $1.6–2.8 million, conditional on the estimator). Counting the moved sales directly gives $2.7 million, collapsed rows $2.4 million, and netting out the control's own excess $1.3 million. Each is under 1% of the $291 million a year that the reform's residential schedule implies on post-reform sales (Table 10).

Table 9: What the 2019 notches cost in avoided tax

Threshold	Jump at X	Extra sales just below X, a year	Missing minus extra, a year	Tax avoided at the jump, $M a year
$1,000,000 (control)	—	102 [17, 180]	−28 [−119, +64]	—
$2,000,000	$5,000	130 [103, 157]	−46 [−103, +17]	0.65 [0.52, 0.79]
$3,000,000	$15,000	54 [34, 72]	+17 [−29, +61]	0.80 [0.51, 1.09]
$5,000,000	$37,500	20 [6, 32]	+7 [−23, +41]	0.74 [0.22, 1.19]
Total, new thresholds				**2.20** [1.63, 2.76]

Notes. Post-reform period, January 2020 to December 2025. Extra sales just below X are the excess mass in $(0.975X, X)$ after the reform minus the pre-reform excess rescaled to the post-reform counterfactual density; missing minus extra is the corresponding missing mass in $(X, 1.05X)$ less that excess. Estimator and windows as in Table 4; 95% bootstrap intervals in brackets, covering sampling uncertainty conditional on the estimator. Each extra sale is assumed to have crossed the threshold from above (Table 2 shows most came from the round price itself) and to avoid the statutory jump at X (at $3,000,000 the buyer's 0.25 points and the seller's 0.25-point base-rate rise); it also lowers the other transfer taxes by about 3–4% of a price reduction that is mostly under $5,000, which is not counted. Other counts give $2.7 million when the moved sales are counted directly as those lost at and above X (Table 2), $2.4 million on collapsed rows, and $1.3 million when each pile is reduced by the control's own excess as a share of its window. The reform's whole residential schedule implies $291 million a year on post-reform sales from $2,000,000 (Table 10). At the $1,000,000 control, whose tax did not change, the extra sales are within the range of the never-taxed placebos. Replication package, `cost_and_sponsors.py`, `focal_inference.py`, `benchmarks_v29.py`.

Transaction volume. Green et al. (2026) estimate that Los Angeles' 4% tax on sales of $5,000,000 or more, 5.5% above $10,000,000, reduced transactions by about half. Nothing of that size is visible here. Sales between $2,000,000 and $10,000,000 were 35.3% of sales between $1,000,000 and $10,000,000 from January 2016 to March 2019 and 34.5% in 2020–2025, a small decline although rising nominal prices alone would push that share up. The 2019 steps were an order of magnitude smaller than the Los Angeles rates, and this comparison does not identify a volume effect; counts over equal three-year periods give the same picture (Supplemental Appendix Table A12).

4.8 A schedule without jumps

The jumps exist because each 2019 rate applies to the whole price. A slice schedule, which charges each rate only on the part of the price above its threshold, has none. Holding recorded prices fixed, Table 10 gives two slice schedules that raise the same $291 million a year from post-reform sales of $2,000,000 or more. One marginal rate of 2.36% on the part above $2,000,000 does it, but it raises the average tax between $2,000,000 and $10,000,000 and lowers it above, so that 77% of these sales would pay more. The statutory steps scaled into marginal rates, from 0.49% above $2,000,000 to 6.12% above $25,000,000, keep the schedule's progressivity: on average they lower the tax in every band below $10,000,000 and raise it above, and 7% of sales would pay more. The remaining kinks change only marginal rates, and the bunching literature generally finds far smaller responses at kinks than at notches (Kleven 2016). The calculation is static.

5 The mortgage recording tax's loan notch

The recording tax is a notched tax on the loan: the borrower's rate rises from 1.80% to 1.925% of the whole loan at $500,000, a jump of $625. Since the threshold did not change in the sample period, the evidence is cross-sectional, and $500,000 is itself the roundest loan amount in this range. The comparison that matters is therefore with other round loan amounts.

Among the 172,725 purchase mortgages with an observed amount, 1,488 are for exactly $500,000 and 1,277 lie in the $1,000 below it. At untaxed round amounts from $300,000 to $700,000, between 79% and 95% of loans in the last $1,000 up to and including the round amount sit at exactly the round amount; at $500,000 only 54% do. Relative to loans in the $25,000 below, loans just under $500,000 are 3 to 10 times as common as loans just under the other round amounts (0.175 against 0.018 to 0.063). The pattern repeats the one in prices: a large share of the loans that would have been written at the round amount are written a few dollars or a few hundred dollars below it, and many are still written at the round amount. Fitted against a smooth counterfactual in the manner of DeFusco and Paciorek (2017), the excess below the threshold is 0.47 (95% interval 0.42 to 0.52), equivalent to $5,900 of loan and a recorded-loan elasticity of 0.059 (0.047 to 0.072), with widths of $5,100 to $6,900 across 27 specifications. Those widths overstate the change in borrowing for the reason Section 4.7 gives for prices.

Two other explanations do not fit. Lenders price loans differently above the conforming loan limit, and the national baseline limit moved from $417,000 in 2016 through $484,350 in

Table 10: The 2019 schedule and two revenue-neutral slice schedules

	2019 notches (whole price)	Slice A (one marginal rate)	Slice B (scaled steps)
Panel A. Tax added by the 2019 reform at a recorded price of			
\$1,999,999	\$0	\$0	\$0
\$2,000,000	\$5,000	\$0	\$0
\$3,000,000	\$22,500	\$23,603	\$4,858
\$5,000,000	\$75,000	\$70,810	\$34,004
\$10,000,000	\$250,000	\$188,825	\$179,733
\$20,000,000	\$600,000	\$424,857	\$689,786
Panel B. Mean tax added, by price band			
\$2–3 million (2,445 a year)	\$6,105	\$10,436	\$2,148
\$3–5 million (1,456 a year)	\$28,657	\$42,981	\$16,822
\$5–10 million (781 a year)	\$101,086	\$111,858	\$84,691
\$10–25 million (236 a year)	\$392,144	\$299,715	\$416,459
\$25 million and above (47 a year)	\$1,354,042	\$967,387	\$2,082,070
Panel C. Totals			
Revenue, \$M a year	291	291	291
Sales paying more than under the notches	—	77%	7%

Notes. Post-reform sales (2020–2025) from \$2,000,000, 4,965 a year after collapsing the repeated rows that multi-lot deeds create (Table 6), with recorded prices held fixed. The 2019 column is the supplemental tax under §1402-b (0.25% from \$2,000,000 to 2.90% from \$25,000,000) and the 0.25-point state base-rate rise at \$3,000,000, each charged on the whole price. A slice schedule charges each marginal rate only on the part of the price above its threshold. Slice A charges 2.36% on the part above \$2,000,000. Slice B applies the statutory increments as marginal rates scaled by a common factor of 1.94, giving cumulative marginal rates of 0.49%, 1.46%, 2.91%, 4.86%, 5.34%, 5.83%, 6.12% above \$2, 3, 5, 10, 15, 20 and 25 million. Both raise the same revenue as the 2019 notches on these sales. A static calculation: it ignores the bunching that removing the jumps would undo and any other behavioral response. Replication package, `benchmarks_v29.py`.

2019 and \$510,400 in 2020 to \$806,500 in 2025; the high-cost limit that applies in New York City was above \$625,000 throughout (Federal Housing Finance Agency 2025). The ratio of loans just below \$500,000 to loans just above stays between 0.53 and 0.97 in every year, wherever the baseline limit stood. And only 5% of the loans just below \$500,000 finance a price within \$1,000 below a transfer-tax threshold, against 2% of those just above, so the pile is not inherited from bunching in prices. The pile is consistent with borrowers responding to the recording tax, by borrowing a little less or moving financing outside the recorded mortgage, which the records cannot tell apart. One untaxed amount, \$550,000, also shows some loans just below it, for reasons not known.

The tax itself is not small. Applied to each observed loan, the borrower-side schedule implies a gross charge of 1.41% of price for houses and 1.28% for condominiums, about \$244 million a year on purchases between \$300,000 and \$6,000,000 in the four ACRIS boroughs,

before credits, exemptions and consolidation adjustments. On the same all-levies basis, that is about half of the residential liability the Department of Finance reports, which also covers refinancings and Staten Island (New York City Department of Finance 2025). A purchase without a recorded mortgage pays none of it. Supplemental Appendix Section I documents the calculation and costs continuous replacement schedules that would raise the same amount from every purchase.

6 What this paper cannot settle

How the moved deals were reached. The records show recorded prices, not the bargaining behind them. The loan check of Section 4.7 rules out a late trim of a price agreed at the round number, but not value moved off the deed into personal property or credits during negotiation, and it cannot say whether a buyer, a seller or both gave up the difference.

Responses beyond the round price. At $3,000,000 and $5,000,000 the response beyond the round price is detectable in the count-based difference-in-differences but not in every bunching specification, and it is not significant net of the control's own change. Its size is therefore uncertain, and the comparison across thresholds has little power once the round-price sales are set aside.

Local responses, so no optimal rate. The widths and elasticities of Table 8 describe responses at discrete jumps, dominated by the round-price margin and by frictions. They do not identify responses to a continuous schedule, and the slice schedules of Table 10 are static. A welfare analysis would also need the economic incidence of each tax, which this paper does not estimate.

Foreclosure transfers are not removed from every sample. Referee's deeds and deeds in lieu of foreclosure are recorded as plain deeds and remain in the sales file, and their consideration can follow a legal rule. They are rare near the 2019 thresholds, and Section 4.6 drops them using a companion paper's classifier (Loschi 2026b) without changing the estimates; that classifier has been checked against AI readings of sampled deed documents, not against a verified record.

7 Conclusion

New York's 2019 reform allows a before-and-after comparison at the same transfer-tax thresholds. At each well-populated new notch, sales moved to just below the threshold, starting with the reform and with nothing similar at never-taxed round prices. Most of the movement is one margin: about 60% of the sales that would have closed at exactly the round-number threshold closed a few thousand dollars below it instead, by less than the buyer's tax saving, and that share does not change between a $5,000 and a $37,500 jump. Beyond the round price the response is modest, detectable at $3,000,000 and $5,000,000 and not at $2,000,000. The avoided tax is $1.3 to $2.7 million a year, under 1% of what the reform's schedule implies.

Two implications follow. For measurement, bunching at a round-number notch should be read against a before period or with the round price treated explicitly: against a smooth counterfactual, sales moved about $2,400 below $2,000,000 register as a pile $16,500 wide, and the same caution applies to loans at the recording tax's $500,000 notch. For design, even a quarter-point notch at a round price moves the deals that would have closed there, whatever the size of the jump. A slice schedule could raise the same revenue without jumps, with the distribution across price bands set by its marginal rates.

Declarations

Data availability. All data are public and free. Sales are New York City Department of Finance annualized calendar sales, NYC Open Data resource `w2pb-icbu`. Deeds and mortgages are ACRIS Real Property Master and Legals, resources `bnx9-e6tj` and `8h5j-fqxa`, linked to the sales in a published dataset (Loschi 2026a), concept DOI `10.5281/zenodo.22977532`. The replication package, concept DOI `10.5281/zenodo.22880953`, holds the analysis data and the code for every table and figure and reproduces them without network access. Preprint versions are under concept DOI `10.5281/zenodo.22925301`.

Declaration of generative AI use. Anthropic's Claude and OpenAI's ChatGPT, including Codex, were used for coding, analysis scripts, reference formatting, drafting and editing, and to cross-check each other's analyses. The author checked the reported results against the deposited outputs and is responsible for the submitted version.

Declaration of competing interest. The author founded HolaCasaClara, a free comparator of Argentine mortgage loans, which has no connection to New York housing records or this study. No other competing interests are reported.

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